

A Fixed-Periodic Bound for Signed Affine Copies of $\{0, 1, 3\}$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

Abstract

Fix an integer $q \geq 2$ coprime to 6 and a nonempty set of residues $R \subseteq \mathbb{Z}/q\mathbb{Z}$. For the truncations of the corresponding periodic set, we compute the limiting density of signed affine copies of $\{0, 1, 3\}$. The limit is at most $1/3$, and equality holds exactly when R is a coset of an additive subgroup. The argument separates a lattice count in the two signed-dilation triangles from a finite closure classification in $\mathbb{Z}/q\mathbb{Z}$. This concerns only the fixed-modulus periodic class; it does not determine the unrestricted finite-set extremal coefficient.

1 Statement and context

For a finite set $A \subset \mathbb{Z}$, write

$$M(A) = \#\{(x, d) \in \mathbb{Z}^2 : d \neq 0, x, x + d, x + 3d \in A\}.$$

Thus both signs of the dilation are counted. This is the signed affine-copy convention used by Korsky [1]. For arbitrary finite sets, that work records the interval coefficient $1/3$ and proves an upper coefficient $47/122$ for the pattern $\{0, 1, 3\}$. Here we address the narrower class obtained by fixing a modulus and a set of allowed residues before taking the size limit.

Cosetwise finite constructions also occur in a different part of finite-group combinatorics. Zeng [2] constructs rainbow trees over finite abelian groups by attaching independently permuted blocks indexed by subgroup cosets and by assembling quotient-indexed fibers. We cite that work only as a methodological neighbor: neither its results nor its rainbow-tree setting imply the affine-copy theorem below, and it is not used in the proof.

Let $G = \mathbb{Z}/q\mathbb{Z}$, let $\emptyset \neq R \subseteq G$, and set

$$A_N = \{1 \leq n \leq N : [n]_q \in R\}.$$

Define the finite modular count

$$T_q(R) = \#\{(u, v) \in G^2 : u, u + v, u + 3v \in R\}.$$

The residue $v = 0$ is included here: it represents all nonzero integer dilations divisible by q , as well as the excluded integer dilation 0. The latter is removed by the signed lattice count below.

Theorem 1. *Let $q \geq 2$ satisfy $\gcd(q, 6) = 1$, and let $\emptyset \neq R \subseteq \mathbb{Z}/q\mathbb{Z}$. Then*

$$\lim_{N \rightarrow \infty} \frac{M(A_N)}{|A_N|^2} = \frac{T_q(R)}{3|R|^2} \leq \frac{1}{3}.$$

Equality holds if and only if R is a coset of an additive subgroup of $\mathbb{Z}/q\mathbb{Z}$.

2 Proof

Put $r = |R|$. Since q is fixed,

$$|A_N| = \frac{r}{q}N + O_q(1). \quad (1)$$

We first count, for each fixed pair of residues $(u, v) \in G^2$, the integer pairs (x, d) in the two possible sign regions.

For $d > 0$, all three points lie in $[1, N]$ exactly when $1 \leq x \leq N - 3d$. Set $m = \lfloor (N - 1)/3 \rfloor$. For a fixed $d \equiv v \pmod{q}$, the number of admissible $x \equiv u \pmod{q}$ is

$$\frac{N - 3d}{q} + O(1).$$

In a fixed residue class modulo q , among $1 \leq d \leq m$ there are $m/q + O_q(1)$ terms, and their sum is $m^2/(2q) + O_q(m)$. Consequently the positive-dilation count is

$$\begin{aligned} C_{u,v}^+(N) &= \frac{1}{q} \sum_{\substack{1 \leq d \leq m \\ d \equiv v \pmod{q}}} (N - 3d) + O_q(N) \\ &= \frac{N^2}{6q^2} + O_q(N). \end{aligned}$$

The condition on the middle point $x + d$ is automatic in this region.

For $d < 0$, write $e = -d \geq 1$. The interval conditions become $1 + 3e \leq x \leq N$, while $x - e$ is then automatically in $[1, N]$. Here $e \equiv -v \pmod{q}$, and the same arithmetic-progression calculation gives

$$C_{u,v}^-(N) = \frac{N^2}{6q^2} + O_q(N).$$

When $v = 0$, these two formulas count $d = q, 2q, \dots$ and $d = -q, -2q, \dots$, respectively. Thus the forbidden integer $d = 0$ belongs to neither triangle and no main-term contribution is lost.

The residue membership of the three points is precisely $u, u + v, u + 3v \in R$. Summing the two estimates over the $T_q(R)$ admissible residue pairs yields

$$M(A_N) = \frac{T_q(R)}{3q^2} N^2 + O_q(N). \quad (2)$$

Equations (1) and (2), together with $r > 0$, prove existence of the asserted limit and its exact formula.

It remains to bound and classify the finite modular count. The bijective change of variables

$$(u, v) \mapsto (a, b) = (u, u + v)$$

rewrites it as

$$T_q(R) = \#\{(a, b) \in R^2 : 3b - 2a \in R\}. \quad (3)$$

Therefore $T_q(R) \leq r^2$. Equality holds exactly when

$$3b - 2a \in R \quad \text{for every } a, b \in R. \quad (4)$$

Assume equality, choose $c \in R$, and put $S = R - c$. Then $0 \in S$, and (4) becomes

$$3y - 2x \in S \quad (x, y \in S).$$

Taking $x = 0$ and $y = 0$ gives $3S \subseteq S$ and $-2S \subseteq S$. Multiplication by 3 and by -2 is injective on G because $\gcd(q, 6) = 1$. Since S is finite, both inclusions are equalities: $3S = S = -2S$. Given $s, t \in S$, choose $y, x \in S$ with $3y = s$ and $-2x = t$. The closure condition gives $s + t = 3y - 2x \in S$. Thus S contains zero and is closed under addition. A finite additive submonoid of a group contains the inverse of each of its elements: if z has order k , then $-z = (k - 1)z$. Hence S is an additive subgroup and $R = c + S$ is its coset.

Conversely, if $R = c + H$ with $H \leq G$, then for $a = c + h_1$ and $b = c + h_2$ one has

$$3b - 2a = c + (3h_2 - 2h_1) \in c + H.$$

Thus every pair in R^2 is counted in (3), so $T_q(R) = r^2$ and the limit is $1/3$. This proves the theorem.

Computational and AI assistance

Generative AI assisted with organizing the proof and preparing the manuscript. The argument was then checked in a separate mathematical review pass. As a supplementary falsification test, an exact finite script checked all nonempty residue subsets for $q = 5, 7, 11, 13$ and several direct finite- N counts. Those computations are not used in the proof.

References

- [1] Samuel Korsky, *Affine Copies of Three-Point Patterns in Sets of Integers*, arXiv:2609.02308v1 [math.NT], 2026.
- [2] Zijian Zeng, *Coset and Fiber Constructions for Rainbow Trees in Finite Abelian Groups*, SSRN eLibrary No. 7387259, 2026, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7387259.